NARAYANA
IIT ACADEMY
INDIA

40+ YEARS OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C) Date:31-12-23
Time: 3HRS Max. Marks: 300

Name of the Student: ____________________ H.T. NO:

31-12-23_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-4_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A galvanometer of resistance $100\,\Omega$ contains 100 division. It gives a deflection of one division on passing a current of $10^{-4}A$. Find the resistance to be connected to it, so that it becomes a voltmeter of range 10V.

A) $\frac{500}{9}\Omega$ B) $500\,\Omega$ C) $\frac{100}{9}\Omega$ D) $900\,\Omega$

2. If mass density of earth varies with distance 'r' from centre of earth as $\rho = kr$ and 'R' is radius of earth, then find the orbital velocity of an object revolving around earth at a distance '2R' from its centre.

A) $\sqrt{\frac{\pi k R^3 G}{4}}$ B) $\sqrt{\frac{\pi k R^3 G}{2}}$ C) $\sqrt{\frac{\pi k R^3 G}{8}}$ D) $\pi k R^3 G$

3. A particle moves clockwise in a circle of radius 1 m with centre at $(x, y) = (1m, 0)$. It starts at rest at the origin at time $t = 0$. Its speed increases at the constant rate of $\left(\frac{\pi}{2}\right) m/s^2$. If the net acceleration at $t = 2\sec$ is $\frac{\pi}{2}\sqrt{\left(1 + N\pi^2\right)}$ then what is the value of N ?

A) 2 B) 4 C) 1 D) 6

4. The rms speed of hydrogen molecules at a certain temperature is 300 m/s. If the temperature is doubled and hydrogen gas dissociates into atomic hydrogen, the rms speed will become

A) 100 m/s B) 300 m/s C) 600 m/s D) $300\sqrt{2}\ m/s$

5. The power of a lens (biconvex) is $1.25\,m^{-1}$ in a particular medium. Refractive index of the lens is 1.5 and radii of curvature are 20 cm and 40 cm respectively. The refractive index of surrounding medium is

A) 1.0 B) $\frac{9}{7}$ C) $\frac{3}{2}$ D) $\frac{4}{3}$

6. Let A_n be the area enclosed by the n[th] orbit in a hydrogen atom. The graph of $\ell n\left(A_n / A_1\right)$ against $\ell n(n)$

A) will not pass through the origin

B) will have certain points lying on a straight line with slope 4

C) will be a monotonically increasing non linear curve

D) will be a circle

7. A screw gauge has 50 divisions on its circular scale. The circular scale is 4 units ahead of the pitch scale marking, prior to use. Upon one complete rotation of the circular scale, a displacement of 0.5mm is noticed on the pitch scale. The nature of zero error involved and the least count of the screw gauge are respectively

A) Positive, 0.1 μm B) Positive, 10 μm C) Negative, 10 μm D) Negative, 0.1 μm

8. In an LC oscillator, if values of inductance and capacitance become twice and eight times, respectively, then the resonant frequency of oscillator becomes $'x'$ times its initial resonance frequency ω_0. The value of $'x'$ is

A) $\frac{1}{4}$ B) 4 C) 16 D) $\frac{1}{16}$

9. A closed organ pipe and an open pipe of same length produce 6 beats per sec when they are set into vibrations simultaneously with their fundamental frequency. If the length of each pipe is doubled, then the number of beats produced is

A) 4 B) 3 C) 5 D) 7

10. A closely wound solenoid of 1000 turns and area of cross – section $2.0 \times 10^{-4}\ m^2$ carries a current of 2.0 A. It is placed with its axis horizontal in a uniform horizontal magnetic field of 0.16 T. If the solenoid is free to turn about the vertical direction. The amount of work (in J) needed to displace the solenoid from its stable orientation to its unstable orientation is $\frac{8x}{250}$. Find the value of x ?

A) 5 B) 4 C) 3 D) 2

11. The ratio of the KE and PE possessed by a body executing SHM when it is at a distance of 1/n of its amplitude from the mean position is

A) n^2 B) $\frac{1}{n^2}$ C) n^2+1 D) n^2-1

12. A fish rising vertically up towards the surface of water with speed $3\,ms^{-1}$ observes a bird diving vertically down towards it with speed $9\,ms^{-1}$. The actual velocity of bird is

(Given $\mu_{water}=\frac{4}{3}$)

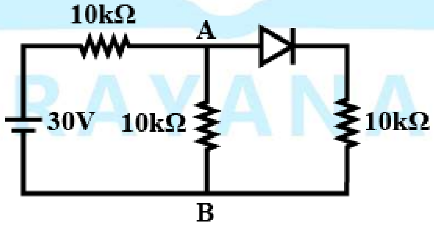
A) $4.5\,ms^{-1}$ B) $5.4\,ms^{-1}$ C) $3.0\,ms^{-1}$ D) $3.4\,ms^{-1}$

13. Electric potential on the surface of a uniformly charged solid sphere is V. Radius of the sphere is R(=1m). Match the following two columns. ('r' is the distance from the centre of the sphere)

Column I		Column II	
(A)	Electric potential at $r=\frac{R}{2}$	(P)	$\frac{V}{4}$
(B)	Electric potential at $r=2R$	(Q)	$\frac{V}{2}$
(C)	Electric field at $r=\frac{3R}{4}$	(R)	$\frac{3V}{4}$
(D)	Electric field at $r=2R$	(S)	$\frac{11V}{8}$

A) A – P; B – R; C – Q; D – S B) A – S; B – P; C – Q; D - R
C) A – S; B – Q; C – R; D – P D) A – R; B – Q; C – P; D – S

14. In the figure, potential difference between A and B is (Diode is ideal)



A) 5 V B) 10 V C) zero D) 15 V

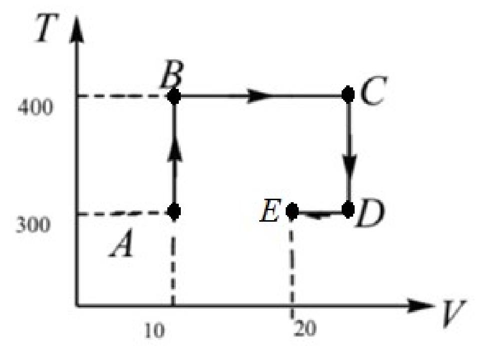
15. In a photoelectric experiment, with light of wavelength λ, the fastest electron has speed v. If the exciting wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will become

A) $v\sqrt{\frac{3}{4}}$ B) $v\sqrt{\frac{4}{3}}$ C) less than $v\sqrt{\frac{3}{4}}$ D) greater than $v\sqrt{\frac{4}{3}}$

16. An insect is at the bottom of a hemispherical ditch of radius 1 m. It crawls, up the ditch but starts slipping after it is at height 'h' from the bottom. If the coefficient of friction between the ground and the insect is 0.75, then 'h' is ($g = 10\,ms^{-2}$)

A) 0.80 m B) 0.60 m C) 0.45 m D) 0.20 m

17. For one mole of mono atomic ideal gas temperature (T in Kelvin) and Volume (V in m^3) relate as shown in the graph. What is the change in internal energy from B to D?
$\left[R = \frac{25}{3} J\,mol^{-1}K^{-1}\right]$



A) 1250 B) 1290 C) 1300 D) 1350

18. Statement 1 : A large soap bubble expands while a small bubble shrinks, when they are connected to each other by a capillary tube

Statement 2 : The excess pressure inside bubble is inversely proportional to its radius

A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement – 1.

B) Statement – 1 is True, Statement – 2 is True; Statement – 2 is not a correct explanation for Statement – 1.

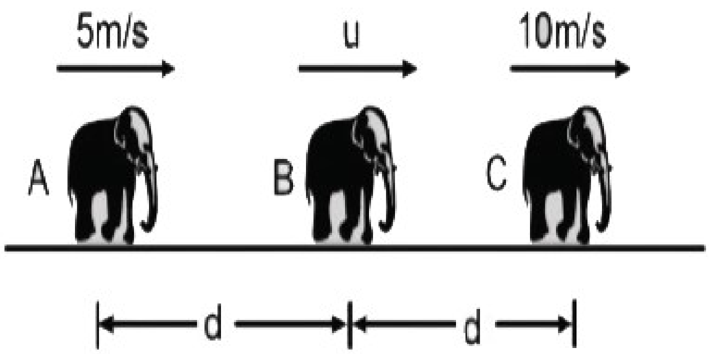
C) Statement -1 is True, Statement – 2 is False.

D) Statement – 1 is False, Statement – 2 is True.

19. Two identical billiard balls are in contact on a smooth table. A third identical ball strikes them symmetrically and comes to rest after impact. The coefficient of restitution is

A) $\frac{2}{3}$ B) $\frac{1}{3}$ C) $\frac{1}{6}$ D) $\frac{\sqrt{3}}{2}$

20. Three elephants A, B and C are moving along a straight line with constant speed in same direction as shown in figure. Speed of A is 5 m/s and speed of C is 10 m/s. Initially separation between A & B is 'd' and between B & C is also d. When 'B' catches 'C' separation between A & C becomes 3d. Find the speed of B (in m/s).



A) 15 B) 20 C) 5 D) 25

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

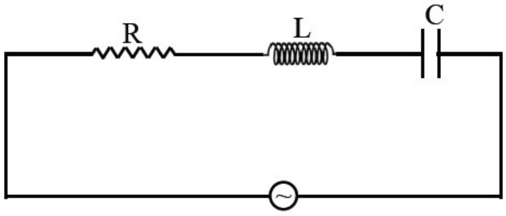
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:
Marking scheme: +4 for correct answer, -1 in all other cases.

21. A beam of natural light falls on a system of 5 polaroids which are arranged in succession such that the pass axis of each Polaroid is turned through 60^0 w.r.t. the preceding one. The fraction of intensity of incident light that passes through the system is $\frac{1}{128\,n}$. Then the value of n ?

22. A 60 pF capacitor is fully charged by a 20 V supply. It is then disconnected from the supply and is connected to another uncharged 60 pF capacitor in parallel. The electrostatic energy that is lost in this process by the time the charge is redistributed between them is (in nJ) ___________

23. A body of mass 2 kg moving with a speed of $4\,ms^{-1}$ makes an elastic collision with another body at rest and continues to move in the original direction but with one fourth of its initial speed. The speed of the centre of mass of two body system after the collision is $\frac{x}{10}ms^{-1}$. Then the value of $'x'$ is __________

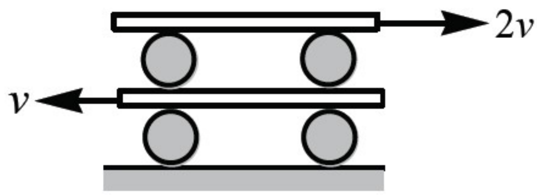
24. In the LCR circuit shown, if RMS voltages across resistor, inductor and capacitor are equal, the RMS value of voltage across resistor will be _________ volt.



$v = 200\sin(100t)$

25. In a cylinder piston arrangement, air is under a pressure P_1. A soap bubble of radius r lies inside the cylinder. Soap bubble has surface tension T. The radius of bubble is to be reduced to half. The pressure P_2 to which air should be compressed isothermally is $AP_1 + \frac{BT}{r}$. Then the value of $A+B$ is ________

26. A system of identical cylinders and plates is shown in figure. All the cylinders are identical and there is no slipping at any contact. The velocity of lower and upper plates are V and 2V, respectively, as shown in figure. Then the ratio of angular speeds of the upper cylinders to lower cylinders is



27. The potential energy (in joule) of a body of mass 2kg moving in the x-y plane is given by $U = 6x + 8y$. Where the position coordinates x and y are measured in metre. If the body is at rest at point (6m, 4m) at time t = 0, it will cross the y-axis at time t equal to

28. An electromagnetic wave of frequency 1×10^{14} hertz is propagating along z-axis. The amplitude of electric field is 4 V/m. If $\epsilon_0 = 8.8\times10^{-12}\ C^2/N-m^2$, then average energy density of electric field will be $N\times10^{-13}\ J/m^3$.

29. The moment of inertia of a uniform circular disc about its diameter is 200 gm cm^2. Then its moment of inertia about an axis passing through its center and perpendicular to its circular face is ______ gm cm^2.

30. A light beam of wavelength 500 nm is incident on a metal having work function of 1.25 eV, placed in a magnetic field of intensity B. The electrons emitted perpendicular to the magnetic field B, with maximum kinetic energy are bent into circular arc of radius 30 cm. The value of B is _____ $\times10^{-7}T$. [Given $hc = 20\times10^{-26}\ J\text{-}m$, mass of electron = $9\times10^{-31}kg$]

CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. In which of the following, mixing of 1N strong acid and 1N strongbaseresults in highest increase in temperature.

A)20 ml acid + 30 ml alkali B)10 ml acid + 40 ml alkali

C)25 ml acid + 25 ml alkali D)35 ml acid + 15 ml alkali

32. Which of the following is an incorrect statement?

A)Half-life of second order reaction decreases with increase in concentration of reactant

B)Half-life of first order is independent of concentration of the reactant

C)The unit of rate constant of zero order is equal to unit of rate

D)The unit of frequency factor A in Arrhenius equation is the unit of half-life of the reaction

33. A set of solution is prepared by using 180g of water as a solvent and 10g of different non-volatile solutes A, B and C. The relative lowering of vapour pressure in the presence of these solutes are in the order [Given, molar mass of A = 100 $g\,mol^{-1}$; B = 200 $g\,mol^{-1}$; C = 10000 $g\,mol^{-1}$]

A)A > B > C B)A > C > B C)C > B > A D)B > C > A

34. Which of the following is an extensive property?

A)Heat capacity B)Refractive index

C)Specific volume D)Molarity

35. A certain metal when irradiated by light $(\upsilon = 3.2\times10^{16}\,Hz)$ emits photoelectrons with twice kinetic energy than the photoelectrons emitted when the same metal is irradiated by light $(\upsilon = 2.0\times10^{16}\,Hz)$. Then υ_0 of metal is

A) $1.2\times10^{14}\,Hz$ B) $8\times10^{15}\,Hz$ (c) $1.5\times10^{16}\,Hz$ D) $4\times10^{12}\,Hz$

36. $KMnO_4$ reacts with KI, in weakly basic medium to form I_2 and MnO_2. When 250 ml of 0.1 M KI solution reacts with 250 mL of 0.02 M $KMnO_4$ in basic medium, what is the number of moles of I_2 formed?

A)0.015 B)0.0075 C)0.005 D)0.01

37. The term invert sugar refers to an equimolar mixture of

A)D-Glucose and D-Galactose B)D-Glucose and D-Fructose

C)D-Glucose and D-Mannose D)D-Glucose and D-ribose

38. Given below are two statements:

Statement-I: Pure Aniline and other arylamines are usually colourless

Statement-II: Arylamines get coloured on storage due to atmospheric reduction

In the light of the above statements, choose the most appropriate answer from the options given below

A)Both statement-I and II are incorrect

B)Both statement-I and II are correct

C)Statement-I is correct but statement-II is incorrect

D)Statement-I is incorrect but statement-II is correct

39. Mark out the incorrect option regarding Cannizzaro reaction

A)The aldehyde, which doesn't have enolisable hydrogen undergo Cannizzaro reaction

B)All types of Cannizzaro reaction e.g. inter, intra and crossed all can be said disproportionation reaction

C)Reaction $PhCHO + CD_2 = O \xrightarrow{OH^-} PhCHD - OH + DCOO^-$

D)Reaction can be first order in base and 2^{nd} order in substrate

40. Identify the alkyne in the following sequence of reactions

$$\text{Alkyne} \xrightarrow[\text{Lindlar's Catalyst}]{H_2} A \xrightarrow[H_2O]{O_3/Zn} \underset{\text{Only}}{B} \xleftarrow[\text{Process}]{\text{Wacker}} H_2C=CH_2$$

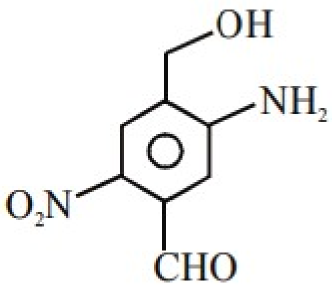
A) $CH_3-C\equiv C-CH_3$

B) $CH_3-CH_2-C\equiv CH$

C) $CH_2=CH-C\equiv CH$

D) $CH\equiv C-CH_2-C\equiv CH$

41. The IUPAC name of the following compound is



A)2-nitro-4-hydroxymethyl-5-amino benzaldehyde

B)3-amino-4-hydroxymethyl-5-nitrobenzaldehyde

C)5-amino-4-hydroxymethyl-2-nitrobenzaldehyde

D)4-amino-2-formyl-5-hydroxymethyl nitrobenzene

42. Lassaigne's test for the detection of nitrogen fails in

A) $NH_2CONHNH_2\cdot HCl$

B) $NH_2NH_2\cdot HCl$

C) NH_2CONH_2

D) $C_6H_5NHNH_2\cdot HCl$

43. Given below are two statements

Assertion (A): Cu^{2+} in water is more stable than Cu^{+}.

Reason (R): Enthalpy of hydration for Cu^{2+} is much more negative than that of Cu^{+}.

In the light of the above statements, choose the correct answer from the options given below

A) Both A and R are correct and R is the correct explanation of A

B) A is correct but R is not correct

C) A is not correct but r is correct

D) Both A and R are correct but R is not the correct explanation of A

44. Which of the following statement is not correct?

A) $B(OH)_3$ is acidic

B)Aqeous solution of Potash alum is acidic in nature

C)The decreasing order of acidic natureof BBr_3, BCl_3 and BF_3 is $BBr_3 > BCl_3 > BF_3$

D) B_2H_6 contains B – B covalent bonds

45. Among the following, the correct statement is

A)Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

B)Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional

C)Between NH_3 and PH_3, NH_3 is a better electron donor because the lone pair of electrons occupies sp^3 orbital and is more directional

D)Between NH_3 and PH_3, PH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

46. Match the list-I with List-II

List-I (Cations)	**List-II (Group reaction)**
(P) Pb^{2+}, Cu^{2+}	(I) H_2S gas in presence of dilute HCl
(Q) Al^{3+}, Fe^{3+}	(II) $(NH_4)_2CO_3$ in presence of NH_4OH
(R) Co^{2+}, Ni^{2+}	(III) NH_4OH in presence of NH_4Cl
(S) Ba^{2+}, Ca^{2+}	(IV) H_2S in presence of NH_4OH.

A)P-I, Q-III, R-II, S-IV B)P-IV, Q-II, R-III, S-I

C)P-III, Q-I, R-IV, S-II D)P-I, Q-III, R-IV, S-II

47. Which of the following is correct stability order of oxides of 17^{th} group elements is

A) $Cl_2 > Br_2 > I_2$ B) $I_2 > Br_2 > Cl_2$

C) $I_2 > Cl_2 > Br_2$ (d) $Cl_2 > I_2 > Br_2$

48. In the electrolysis of $CuCl_2$ solution using Cu electrodes, the weight of Cu increased by 2 gram at cathode.At the anode:

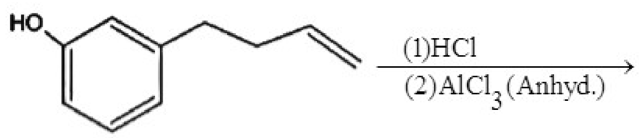
A)0.2 mole of Cu^{2+} will go into solution

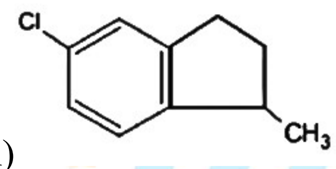
B)560 ml of O_2 liberate

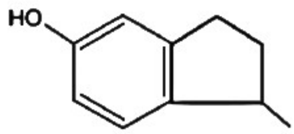
C)No loss in weight

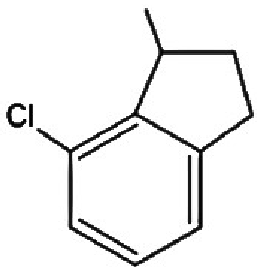
D)2 gram of copper goes into solution as Cu^{2+}

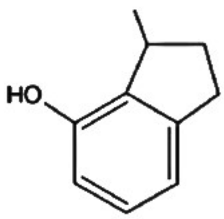
49. The major product of the following reaction is



A) 

B) 

C) 

D) 

50. The conductivity measurement of a coordination compound of Cobalt (III) shows that it dissociates into 3 ions in solution. The compound is

A)Hexaamminecobalt(III) chloride

B)Pentaamminesulphatocobalt(III) chloride

C)Pentaamminechloridocobalt(III) sulphate

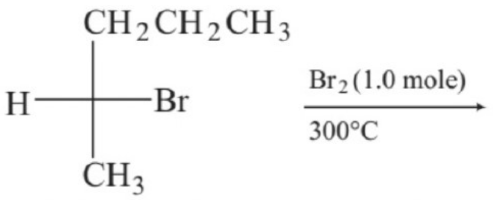
D)Pentaaminechloridocobalt (III) chloride

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

51. In the following monobromination reaction, the number of possible chiral product(s) is/are ____



(1.0 mole) (Enantiomerically pure)

52. The number of geometrical isomers possible for the complex $[Co\,L_2Cl_2]^-(L = H_2NCH_2CH_2O^-)$ is

53. Consider the following reaction $6NaOH + 3Cl_2 \longrightarrow 5NaCl + A + 3H_2O$. What is the oxidation number of chlorine in "A"?

54. On complete combustion, 0.492g of an organic compound gave 0.792 g of CO_2. Then the % of carbon in the organic compound is ____________ (Nearest Integer)

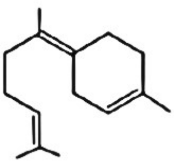
55. The number of $d\pi - p\pi$ bonds in ClO_4^-

56. The value of $\log_{10} K$ for a reaction $A \rightleftharpoons B$ is (Given: $\Delta_r H^0_{298K} = -54.07\,kJ\,mol^{-1}$, $\Delta_r S^0_{298K} = 10\,JK^{-1}\,mol^{-1}$ and $R = 8.314\,JK^{-1}mol^{-1}$, $2.303 \times 8.314 \times 298 = 5705$)

57. A decapeptide (Molecular weight 796) on complete hydrolysis gives glycine (Molecular weight 75), alanine and phenylalanine. Glycine contributes 47.0% to the total weight of the hydrolyzed products. The number of glycine units present in the decapeptide is

58. The total number of cyclic isomers possible for a hydrocarbon with the molecular formula C_4H_6 is

59. Consider the reaction: $AB_2(g) \rightleftharpoons AB(g) + B(g)$. If the initial pressure of AB_2 is 500 torr and equilibrium pressure is 600 torr, equilibrium constant K_p in terms of torr is

60. Reductive ozonolysis of a terpenoid of following structure  gives how many different products.

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. Let $f(x)=\dfrac{x^2\sin\left(\frac{1}{x}\right)+2x}{(1+x)^{\frac{1}{x}}-e}$ then $\lim\limits_{x\to 0} f(x)=$

A) $\frac{-2}{e}$ B) $\frac{2}{e}$ C) $\frac{4}{e}$ D) $\frac{-4}{e}$

62. In a triangle ABC, BC= 3 and AC= 4 and circle with AB as diameter passes through the centroid of a triangle, then AB is

A) $\sqrt{5}$ B) 5 C) $\sqrt{3}$ D) $\frac{5}{2}$

63. The value of definite integral $\int_{-\pi}^{\pi}\frac{2x(1+\sin x)}{1+\cos^2 x}dx$ is:

A) $\frac{\pi}{2}$ B) π C) π^2 D) $\frac{\pi^2}{2}$

64. Let $f(x)=x^3+x+1$ and $g(x)$ be its inverse then $g'(3)=$__________.

A) $\frac{1}{4}$ B) $-\frac{1}{4}$ C) 4 D) –4

65. Let $f(x)$ be a cubic polynomial on R which increases in the interval $(-\infty,0)$ **and in** $(1,\infty)$ and decreases in the interval $(0,1)$. If $f'(2)=6$ and $f(2)=2$, then the value of $\tan^{-1}(f(1))+\tan^{-1}\left(f\left(\frac{3}{2}\right)\right)+\tan^{-1}(f(0))$ is equal to:

A) $\tan^{-1}2$ B) $\cot^{-1}2$ C) $-\tan^{-1}2$ D) $-\cot^{-1}2$

66. Let the equations of two adjacent sides of a parallelogram ABCD be $2x-3y=-23$ and $5x+4y=23$. If the equation of its one diagonal AC is $3x+7y=23$ and the distance of A from the other diagonal is d, then $50d^2$ is equal to__________

A) 501 B) 529 C) 468 D) 731

67. A survey of 500 television watchers produced the following information, 285 watch football, 195 watch hockey, 115 watch basketball, 45 watch football and basketball, 70 watch football and hockey, 50 watch hockey and basketball, 50 do not watch any of the three games. How many watch all the three games? How many watch exactly one of the three games?

A) 190 B) 40 C) 25 D) 325

68. The mirror image of the curve $\arg\left(\frac{z+i}{z-1}\right)=\frac{\pi}{4}$, $i=\sqrt{-1}$ in the line $z(i-1)+\bar{z}(1+i)=0$, **in argand plane , is**

A) $\arg\left(\frac{z+i}{z+1}\right)=\frac{\pi}{4}$ B) $\arg\left(\frac{z+1}{z-i}\right)=\frac{\pi}{4}$

C) $\arg\left(\frac{z-i}{z+1}\right)=\frac{\pi}{4}$ D) $\arg\left(\frac{z+i}{z-1}\right)=\frac{\pi}{4}$

69. If $x=\sin^{-1}(\sin 10)$ and $y=\cos^{-1}(\cos 10)$, then $y-x$ is equal to:

A) π B) 0 C) 10 D) 7π

70. Statement 1:Length of side of an equilateral triangle is 24. The mid points of the sides are joined to form another triangle whose midpoints of sides are joined to form another triangle and continue the process infinite number times. Then sum of perimeters of all such triangles formed is 144.

Statement 2 : If $\log_2(a+b)+\log_2(c+d)\geq 4$ then the minimum value of a+b+c+d is 8

A) Statements 1 & 2 both are correct

B) Statements 1 & 2 both are false

C) Statement 1 is true, statement 2 is false

D) Statement 1 is false, statement 2 is true

71. For each real x, -1<x<1. Let A(x) be the matrix $(1-x)^{-1}\begin{bmatrix}1 & -x\\ -x & 1\end{bmatrix}$ $and\ z=\frac{x+y}{1+xy}$. Then

A) $A(z)=A(x)A(y)$ B) $A(z)=A(x)-A(y)$

C) $A(z)=A(x)+A(y)$ D) $A(z)=A(x)[A(y)]^{-1}$

72. If α, β are the roots of the equation $\lambda\left(x^2 - x\right) + x + 5 = 0$. If λ_1 and λ_2 are two values of λ for which the roots α, β are related by $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{4}{5}$, then the value of $\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1}$, is

A) 254 B) 200 C) 350 D) 154

73. Let $A = \{1,2,3,4,5,6,7\}$. The number of surjective functions defined from A to A such that $f(i) = i$ for atleast four values of i from $i = 1,2,....,7$, is:

A) 7! B) 92 C) 126 D) 407

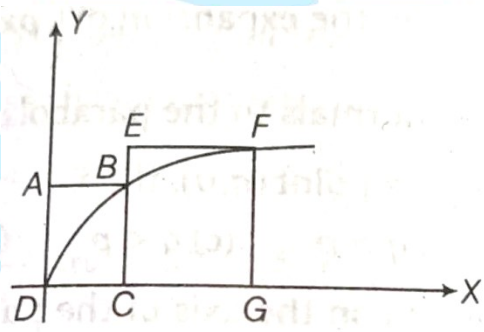
74. Let a line l pass through the origin and be perpendicular to the lines

$l_1 : \vec{r} = \left(\hat{i} - 11\hat{j} - 7\hat{k}\right) + \lambda\left(\hat{i} + 2\hat{j} + 3\hat{k}\right), \lambda \in R$ and $l_2 : \vec{r} = \left(-\hat{i} + \hat{k}\right) + \mu\left(2\hat{i} + 2\hat{j} + \hat{k}\right), \mu \in R$.

If P is the point of intersection of l and l_1, and $Q(\alpha, \beta, \gamma)$ is the foot of perpendicular from P on l_2, then $(\alpha + \beta + \gamma)$ is ________

A) $\frac{5}{9}$ B) $\frac{9}{5}$ C) $\frac{19}{5}$ D) $\frac{5}{3}$

75. ABCD and EFGC are squares and the curve $y = \lambda\sqrt{x}$ passes through the origin D and the points B and F. The ratio $\frac{FG}{BC}$ is (as shown in the figure).



A) $\frac{\sqrt{3}+1}{4}$ B) $\frac{\sqrt{3}+1}{2}$ C) $\frac{\sqrt{5}+1}{4}$ D) $\frac{\sqrt{5}+1}{2}$

76. The value of the expression $\frac{\sin^2 \frac{2\pi}{7}}{\sin^2 \frac{\pi}{7}} + \frac{\sin^2 \frac{4\pi}{7}}{\sin^2 \frac{2\pi}{7}} + \frac{\sin^2 \frac{\pi}{7}}{\sin^2 \frac{4\pi}{7}}$ is equal to:

A) 3 B) 4 C) 5 D) 6

77. Two cards are drawn successively with replacement from a well shuffled deck of 52 cards, then the mean of the number of aces is

A) $\frac{1}{13}$ B) $\frac{3}{13}$ C) $\frac{2}{13}$ D) $\frac{1}{2}$

78. STATEMENT-1: The integral part of $\left(8+3\sqrt{7}\right)^{20}$ is an even integer.

STATEMENT–2: $\left(8+3\sqrt{7}\right)^{20}+\left(8-3\sqrt{7}\right)^{20}$ is an even integer.

A) Statements 1 & 2 both are correct

B) Statements 1 & 2 both are false

C) Statement 1 is true, statement 2 is false

D) Statement 1 is false, statement 2 is true

79. If an unbiased die, marked with –2,–1,0,1,2,3 on its faces, is thrown five times, then the probability that the product of the outcomes is positive, is:

A) $\frac{881}{2592}$ B) $\frac{521}{2592}$ C) $\frac{440}{2592}$ D) $\frac{27}{288}$

80. The following data gives the distribution of height of students:

Height (in cm)	160	150	152	161	156	154	155
Number of students	12	8	4	4	3	3	7

The median of the distribution is

A) 154 B) 155 C) 160 D) 161

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. A function $f:R\to R$ satisfies the equation $f(x)f(y)-f(xy)=x+y,\forall x,y\in R$ and $f(1)>0$, then $f(2)f^{-1}(2)=$_________

82. For y > 0 and $x\in R, ydx+y^2dy=xdy$ where $y=f(x)$. If $f(1)=1$, then the value of $f(-3)$ is

83. Let $f(x)$ be a polynomial of degree 3. If the curve $y=f(x)$ has relative extrema at $x=\frac{\pm 2}{\sqrt{3}}$ and passes through $(0,0)$ and $(1,-2)$ dividing the circle $x^2+y^2=4$ in two parts, then the area bounded by $x^2+y^2=4$ and $y\ge f(x)$ is $\frac{k\pi}{2}$. Find the value of k.

84. Suppose $\int \frac{1-7\cos^2 x}{\sin^7 x\cos^2 x}dx=\frac{g(x)}{\sin^7 x}+C$, where C is an arbitrary constant of integration. The find the value of $g'(0)+g''\left(\frac{\pi}{4}\right)$.

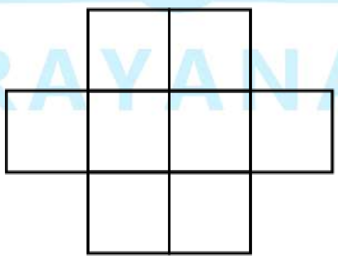
85. The urns A,B and C contain 4 red, 6 black; 5 red, 5 black and λ red, 4 black balls respectively. One of the urns is selected at random and a ball is drawn. If the ball drawn is red and the probability that it is drawn from urn C is 0.4 then $\lambda =$ ____________.

86. ABCD is a parallelogram. L is point on BC which divides BC in the ratio 1:2 AL intersects BD at P. M is a point on DC which divides DC in the ratio 1:2 and AM intersects BD in Q, then $\frac{BD}{PQ}=$ __________.

87. From a given solid cone of height H, another inverted cone whose axis coincides with the axis of the given cone, is carved whose height is h, such that it's volume is maximum, then the ratio $\frac{H}{h}=$ ____________.

88. The radius of the circle passing through the points of intersection of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $x^2-y^2=0$ is k, then [k] = ______________([.] is G.I.F)

89. Six X's have to be placed in the squares of the figure below, such that each row contains atleast one X. In how many different ways can this be done?



90. Given the matrix $A=\begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$. If $xyz=60$, $8x+4y+3z=20$ and $A(adjA)$ is equal to kI, ($k\in R$, I is identity matrices of order 3) then k = _______________